Template Week 3 – Hardware

Student number:510586

Assignment 3.1: Examine your phone

What processor is in your phone?

The processor in my phone is the SnapDragon 8 elite

To which architecture family does this processor belong? In other words, which Instruction Set Architecture (ISA) is used?

ARMv9.2-a

How much RAM is in it?

I have 12 GB of RAM

How much storage does your phone have?

264GB

What operating system is running on your phone?

android

Approximately how many applications do you have installed?

25

Which application do you use the most?

youtube

Can your phone be charged with what type of plug?

USB-C

Which I/O ports can you visually see on your phone?

My USB-C

Assignment 3.2: Examine your laptop

What processor is in your laptop?

Intel Core i7-12650H

To which architecture family does this processor belong? In other words, which Instruction Set Architecture (ISA) is used?

X86-64

How much RAM is in it?

16GB of RAM

How much storage does your laptop have?

512GB

Which operating system is running on your laptop?

Windows

Approximately how many applications do you have installed?

15

Which application do you use the most?

Steam

Can your laptop be charged with what type of plug?

It uses a Barrel plug

Which I/O ports can you visually see on your laptop?

3 USB-A 2.0, 1 USB-C, mic in, aux in, ethernet , HDMI

Assignment 3.3: Power to the laptop

What is the input voltage?

100-240v

What is the output voltage?

20v dc

How many watts can your power adapter deliver?

120W

Is the input voltage AC or DC?

AC

Is the output voltage AC or DC?

DC

AC/DC what is that?

AC is alternating current means the current can change directions

DC is Direct current means the current goes only in one direction.

If you reverse the polarity of the output voltage, is that bad for your laptop?

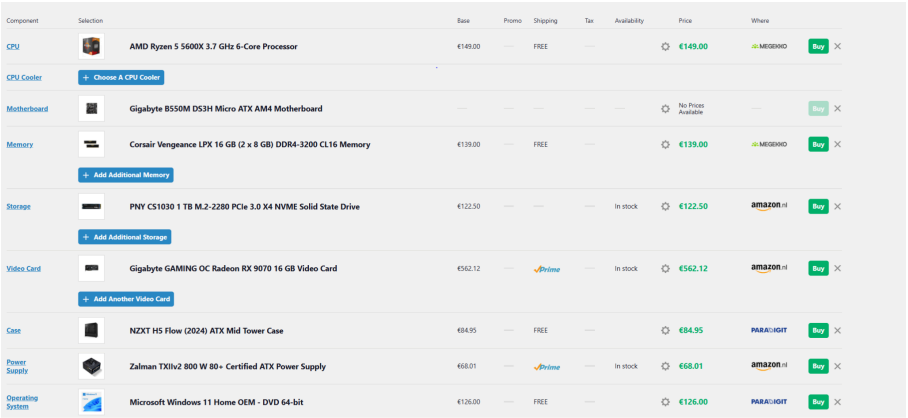
yes , could short circuit my laptop

You forgot your power adapter, your laptop normally needs 15 watts. You will be loaned a power adapter that can deliver 50 watts. Voltage, polarity, etc. are all the same compared to the original power adapter. You can connect the borrowed power adapter to your laptop. What will happen? Also explain why you think that.

If the only thing that changes is the amount of watts the charger supplies then it will charge my pc without any issues,as long as the voltage and polarities match then the 35 watts that are over 15 will just go unused.

Assignment 3.4: Build your dream PC

Screenshots PC configuration + motivation:



Component	Selection	Base	Promo	Shipping	Tax	Availability	Price	Where	
CPU	AMD Ryzen 5 5600X 3.7 GHz 6-Core Processor	€149.00	—	FREE	—		€149.00	MEGEKKO	Buy
CPU Cooler	+ Choose A CPU Cooler								
Motherboard	Gigabyte B550M DS3H Micro ATX AM4 Motherboard	—	—	—	—	—	No Prices Available	—	Buy
Memory	Corsair Vengeance LPX 16 GB (2 x 8 GB) DDR4-3200 CL16 Memory	€139.00	—	FREE	—		€139.00	MEGEKKO	Buy
	+ Add Additional Memory								
Storage	PNY CS1030 1 TB M.2-2280 PCIe 3.0 X4 NVME Solid State Drive	€122.50	—	—	—	In stock	€122.50	amazon.nl	Buy
	+ Add Additional Storage								
Video Card	Gigabyte GAMING OC Radeon RX 9070 16 GB Video Card	€562.12	—	Prime	—	In stock	€562.12	amazon.nl	Buy
	+ Add Another Video Card								
Case	NZXT H5 Flow (2024) ATX Mid Tower Case	€84.95	—	FREE	—		€84.95	PARADIGIT	Buy
Power Supply	Zalman TXIIv2 800 W 80+ Certified ATX Power Supply	€68.01	—	Prime	—	In stock	€68.01	amazon.nl	Buy
Operating System	Microsoft Windows 11 Home OEM - DVD 64-bit	€126.00	—	FREE	—		€126.00	PARADIGIT	Buy

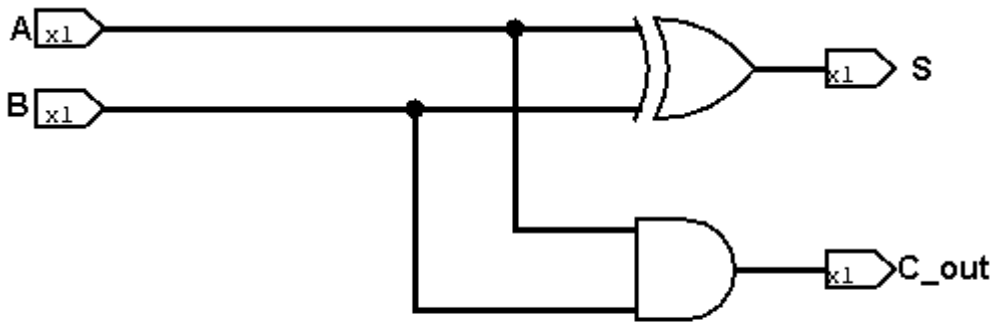
This specific setup is what I have at home, it was built so I am able to play games and do some very light photo/video editing and with home headroom for the future if I want to 3d modelling.

The Main difference between my home setup and my laptop is the overall power of the processor and the graphics card i have in my home desktop.

Assignment 3.5: Adders

Complete the **half adder**, **full adder** and **4-bit adder** assignment as described in the PowerPoint slides of week 3 in Logisim. Save the chip design and also export three PNG pictures of the separate finished designs. See the PowerPoint slides of week 3.

Paste the three exported PNG pictures in here.



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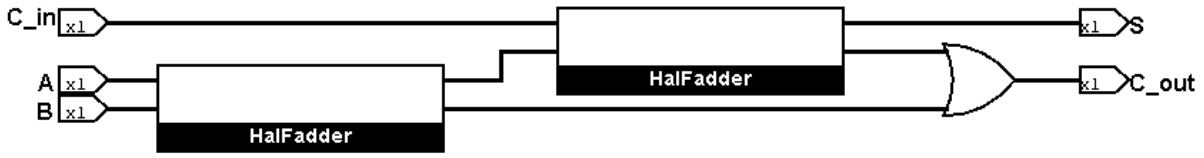

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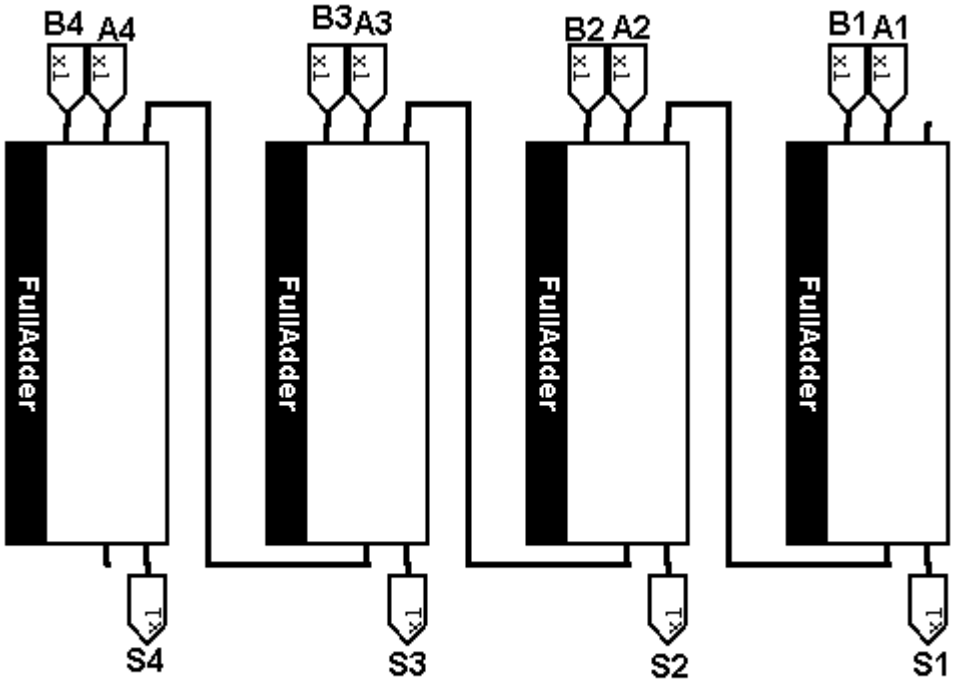

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